

Supplementary Materials for
Slow-light silicon modulator with 110-GHz bandwidth

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Sci. Adv. **9**, eadi5339 (2023)
DOI: 10.1126/sciadv.adi5339

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1 The coupling coefficient and band structure of cascaded corrugated waveguide

To elaborate on the relationship and constraint between a set of modulator parameters, we start by defining the coupling coefficient $\kappa = \int d^3\mathbf{r}[\varepsilon_0(\mathbf{r}) - \varepsilon(\mathbf{r})]\mathbf{E}_\Omega(\mathbf{r}) \cdot \mathbf{E}_\Omega(\mathbf{r} - \mathbf{R})$ with R representing the distance between the adjacent resonators. Since the electric field E_Ω of the mid-gap mode decays exponentially along the grating, so we express $E_\Omega = E_0 e^{-\Delta \cdot x}$, in which Δ is the depth of the one-dimensional photonic crystal bandgap. According to the CROW theory, the electric field E_Ω should be normalized, and thus $|E_0|^2$ is inversely proportional to the grating length of N_p . Then we simplify the coupling coefficient κ of the mid-gap mode as:

$$\begin{aligned}\kappa &= \int d^3\mathbf{r}[\varepsilon_0(\mathbf{r}) - \varepsilon(\mathbf{r})]\mathbf{E}_\Omega(\mathbf{r}) \cdot \mathbf{E}_\Omega(\mathbf{r} - 2N_p a \mathbf{e}_x) \\ &= 2\kappa_1(ff)\Delta\varepsilon E_0^2 \int_0^\infty e^{-\Delta \cdot x} \cdot e^{-\Delta \cdot (x+2N_p a)} dx \\ &= \kappa_0 e^{-2N_p a \Delta} / N_p,\end{aligned}\tag{S1}$$

where $\kappa_1(ff)$ stands for the overlapping coefficient in a unit cell which is related to the filling factor of grating, and the electric field distribution of the mode; N_p denotes the number-of-period of grating at one side of the phase shifter, and κ_0 is a constant over N_p or N_r , but depends on the geometry of the unit cell and inversely proportional to the bandgap depth of Δ . Accordingly, in the case that the coupling is sufficiently weak, we only take into account the adjacent resonators, and the dispersion relation of the supercell band ω_K can be expressed as^{59,60}

$$\omega_K = \Omega[1 - \chi/2 + \kappa \cos(KL_0)],\tag{S2}$$

in which Ω presents the angular frequency of the isolated mode, $\chi/2$ represents the detuning from resonator cascading, and L_0 stands for the length of one resonator. Equation S2 indicates that the

inter-resonator coupling turns an isolated mode into a continuous supercell band, which is shown in Fig. 1E.

2 Discussion on the nonlinear dependency of modulation efficiency on the voltage

Modulation efficiency is a critical parameter of the electro-optic modulator, which is usually characterized by the half-wave voltage or V_π , referred to as the voltage required for phase change π . Such a convention was widely adopted in lithium niobate modulators literature^{27,28}, in which the modulation efficiency is linearly proportional to the driving voltage. However, for a slow-light silicon modulator based on plasma dispersion, the modulation efficiency (EF) nonlinearly depends on the voltage, given by:

$$EF = \frac{\phi}{\pi V} = \eta_0 \cdot N_p N_r \cdot \frac{e^{2N_p a \cdot \Delta}}{\pi \kappa_1} / (\sqrt{V + V_{bi}} + \sqrt{V_{bi}}). \quad (S3)$$

According to Equation. S3, the linear dependence of modulation efficiency on the voltage only holds at the approximation of small signal modulation, and the differential efficiency relies on the bias voltage too. If one tries to estimate the half-wave voltage from a small signal, some complexities should be taken into account. Specifically, since it isn't easy to detect the phase of light in optics directly, the detection of phase is usually transformed into the detection of light intensity. The output intensity I for MZI is a cosine function of the phase difference ϕ , which should be addressed. More importantly, because the accumulated phase is nonlinear to the reverse bias, an incredibly higher voltage is required to achieve a full-range phase of π for a slow light silicon modulator, which is neither feasible nor necessary in the measurement and application

because a functional modulator only asks for sufficient modulation depth but not full-range of π to accomplish data transmission. Therefore, we use the merit of modulation efficiency at a specific bias voltage rather than half-wave voltage to best capture the effectiveness of EO modulation and calculate Fig. 2B,E in the main text by using Equation S3.

3 Estimation of Q s and modulation efficiency of the slow light modulator

As discussed in the main text, Q_{loading} represents the energy leakage at the ends of the slow waveguide. Here we estimate Q_{loading} by comparing the energy stored inside the resonator to the leakage at the end facets. For a single resonator, we have:

$$\begin{aligned}
 Q_{\text{loading}} &= 2\pi c / \lambda_0 \frac{W}{P_{\text{loading}}} \\
 &= 2\pi c / \lambda_0 \frac{\int_0^{N_p a} \frac{1}{2} \varepsilon (E_0 e^{-\Delta \cdot x})^2 dx \cdot S}{\frac{1}{2} \varepsilon c (E_0 e^{-\Delta \cdot N_p a})^2 \cdot S} \\
 &= \frac{\pi}{\lambda_\varepsilon \cdot \Delta} (e^{2N_p a \cdot \Delta} - 1) \\
 &= \frac{\pi}{2a \cdot \Delta} (e^{2N_p a \cdot \Delta} - 1),
 \end{aligned} \tag{S4}$$

where λ_ε is the wavelength inside the dielectric medium, which is around $2a$. Furthermore, when multiple resonators cascade together, P_{loading} maintains the same as to the single resonator, but W grows proportional to the number-of-resonator N_r . Therefore, for the whole device with N_r of resonators, its Q can be estimated as $Q_{\text{loading}}(N_r) = N_r Q_{\text{loading}}(N_r = 1)$.

We further estimate the modulation efficiency of the slow light waveguide from the carrier dispersion.

As mentioned in the main text, the phase accumulation is proportional to $\sqrt{V}$, which actually relies on the carrier dispersion of the PN junction. Specifically, when a reverse bias voltage V is applied to the PN junction, the width of the depletion region is given by:

$$w_{\text{dep}} = \sqrt{\frac{2\epsilon_r\epsilon_0}{e} \frac{N_A + N_D}{N_A N_D} (V + V_{bi})}, \quad (\text{S5})$$

where V_{bi} is the build-in voltage; N_A and N_D stands for the doping density. For our design, at a doping density of $5.0 \times 10^{17}/\text{cm}^3$, and reverse bias V of 10 V, the depletion width w_{dep} is estimated as 239 nm, which is only in half core-width of the waveguide. Considering the reverse bias we applied in the experiment, we are expecting a simply linear dependency between the effective field overlapping (the carriers and optical mode) and the width of depletion region. Namely, we have $\eta \propto w \propto (\sqrt{V + V_{bi}} - \sqrt{V_{bi}})$, which is applied to estimate modulation phase ϕ in the main text and modulation efficiency EF in Equation S3.

4 The impact and limitation of electrode length

Rather than the RC constants of PN junctions, we found that the length of the electrode brings a serve bottleneck that forbid the modulator to operate at ultra-high speed. As reported in the literature, long silicon modulators are usually difficult in achieving high bandwidth. The impact of electrode lengths comes from two aspects: (1) the microwave loss of the electrode itself; (2) the group velocity mismatching between the microwave and lightwave.

To quantitatively evaluate the impact of electrode length, we simulate the EE S_{21} responses of electrodes by using HFSS software. As shown in Fig. S2, we calculate the S_{21} curves from

1 GHz to 300 GHz, for a series of electrode lengths from 100 μm to 500 μm . Clearly, the S_{21} responses are sensitive to the electrode lengths, proved by the increment of RF loss with increasing lengths. Although the trend is qualitatively correct, the RF losses are considerably small since we applied an ideal metal electrode model in the simulation. To best capture the impact of electrodes in the experiment, we model the RF losses from previous measurements (Fig. S2C) in which the EE S_{21} responses of electrodes in the lengths of 2 mm, 3 mm, and 4 mm are illustrated. Besides, to rule out the influence of PN junction that should be in principle a constant value upon device lengths, we calculate the differential S_{21} responses and normalize to a length of 100 μm to reflect the loss of the 1 mm electrode itself (Fig. S2D). By updating the electrode model with practical RF losses, we re-calculate the EE S_{21} response as shown in Fig. S2B (the same as Fig. 2G in the main text), indicating that the RF losses are much large than the ideal case and would set an upper bound of electrode length in a practical device.

On the other hand, the mismatching between the microwave and lightwave velocity also limits the operational bandwidth and thus lengths of the electrodes, governed by⁶⁵:

$$f_{3\text{dB}} = \frac{0.18}{l(n_m - n_g)} (\text{GHz}) \quad (\text{S6})$$

in which l is the electrode length; n_m and n_g are the group index of microwave and lightwave, respectively. We estimate $n_m \sim 2.1$ at 110 GHz. By using such parameters, we calculate the mismatching-limited bandwidth as shown in Fig. S3 for a series of electrode lengths from 100 μm to 500 μm and group index n_g from 4 to 12. Our optimal design of $n_g = 6.1$ is highlighted as star marks. The results indicate that it is important to select a suitable group index that coordinates

with an appropriate electrode length for better microwave-light matching. Short length, or too weak (even the absence of) slow light effect would lead to insufficient phase accumulation of the modulator, while too long, or too strong slow light effect would cause a large microwave-light mismatch. Our optimal design indeed works at comprehensive balanced conditions.

5 Optical loss of the slow light waveguide

As a proof-of-concept design, we didn't optimize the transition region that adiabatically connects the conventional waveguide and the slow light waveguide, and therefore, there exists some abrupt change of group velocities between the conventional waveguide mode to the slow light mode, thus will bring some additional reflection loss. In the comparison test, the total insertion loss of Device A with 124 μm length ($N_p = 20$, $N_r = 10$) is measured to be 6.8 dB, while that of Device B with 249 μm length ($N_p = 20$, $N_r = 20$) is 10.5 dB. Thus, the practical loss of the slow light waveguide with $N_r = 10$ is 3.7 dB, and the loss per unit length of the slow light waveguide is calculated as 29.8 dB/mm. The waveguide loss can be reduced by improving the fabrication process, such as reducing the side roughness, so as to enhance the device performance further.

6 Modulation efficiency of the proposed modulator

In the main text, we applied the efficiency factor rather than the conventional used V_π and $V_\pi L$ to depict the modulation efficiency, mainly because the modulation efficiency in the proposed modulator exhibits quite a nonlinear response with the applied voltage. Indeed, compared to the conventional modulators, the proposed modulator requires a much higher voltage range to achieve

a full-range phase shift of π , but it is usually not necessary in practice since a just sufficient modulation depth (not full-range) can create enough optical extinction ratio to support the datacom applications.

Nevertheless, for the convenience of comparing with existing modulators, we measure the half-wave voltage of the proposed modulator through the dependence between transmitted optical power and applied voltage by using a fitting method (Fig. S 4). From the experimental results, we found the total normalized transmission variation is about 10% under the voltage range from 0 V to 5 V, corresponding to a phase change of 0.1π . We further perform a sinusoidal fitting of optical transmission with the voltage around the bias point, resulting in a half-wave voltage of 78 V based on a length of 124 μm . Accordingly, the modulation efficiency is obtained as 0.96 V $\cdot\text{cm}$ experimentally, which is a relatively high value among silicon modulators. Combining with the propagation loss α (29.8 dB/mm), the $\alpha V_{\pi} L$ is calculated as 286.1 dB $\cdot\text{V}$. Noteworthy is that the modulation efficiency can be further optimized. For instance, we can dope more heavily and optimize coupling loss at the both input and output ends, which can preserve the overall loss level while promote the modulation efficiency.

7 Spectra characteristics of the slow light modulator

The spectra measurement setup is illustrated in Fig. S5. A tunable laser generates continuous wave (CW) light in the spectral range from 1460 nm to 1640 nm. A polarization controller (PC) is used to adjust the polarization state. The light is then sent to the modulator via the grating coupler. The

alignment between the fiber and the grating coupler is realized by adjusting the position between them through a six-dimensional alignment toolkit. The bias voltage is applied to the modulator through a high-frequency probe. The light is collected from the output grating coupler by using a lensed fiber and then send to the optical spectrum analyzer (OSA) for further analysis.

The transmission spectra of the three devices are shown in Fig. S6, which are Device A ($N_p = 20$, $N_r = 10$), Device B ($N_p = 20$, $N_r = 20$), and Device C ($N_p = 40$, $N_r = 10$), respectively. As shown in Fig. S6A to D, the overall spectra of Device A and B look similar since they have the same number-of-period (N_p) which is consistent with the theory. But the passbands of the two devices shrink from 8 nm (Device A) to 7 nm (Device B) because the number-of-resonator (N_r) is doubled thus weakening the tunneling effect and bringing a smaller coupling coefficient κ accordingly. The spectra of Device C changes greatly compared with that of Device A and B, and the passband width dramatically narrows. Although the total device length of Devices A and C are quite close, they behave differently in the spectrum which would result in disparate modulation performance. The measured spectra under different N_p and N_r are in great consistency with the theory.

We note that the difference in the spectral response between theoretical and experimental results is mainly contributed by the bandwidth limit of the input and output grating couplers. The spectral response of the grating couplers is not completely uniform, which results in an extra modification of the observed output intensity. To quantitatively characterize the grating couplers, we measure the transmission spectra of six pairs of input and output grating couplers that are

connected via a short strip waveguide (300 μm). The measured spectral profiles are consistent with that of in Fig. S7, showing that the uneven output spectrum of the modulators is mainly due to the grating couplers.

8 The measurement of EE and EO bandwidths

The setup of electrical-to-electrical (EE) bandwidth measurement is illustrated in Fig. S8. The S-parameter is measured by using the vector network analyzer (VNA). Briefly, the VNA generates a high-frequency RF signal from Port 1, which is mixed with the DC bias voltage through Bias-Tee and loaded onto the modulator. The RF signal is applied to the modulator through the pads of the electrodes, and then output from the pads on the other side of the electrodes. Further, the signal is transmitted to Port 2 of the VNA, to compose a closed loop for the S parameter and EE bandwidth measurement. SOLT calibration is performed before the measurement to eliminate the influence of irrelevant variables in the system.

When the impedance and velocity are perfectly matched, the RF attenuation is the main factor to limit the operation bandwidth of the modulator. In this case, electrical-optical (EO) 3-dB bandwidth is reached at a certain frequency when EE bandwidth is around 6.4 dB¹⁷. Therefore, the EE bandwidth can serve as a criterion for evaluating of EO performance. The experimental results are shown in Fig. S9, in which the RF attenuation of modulators with different design parameters are plotted. Noticed that an increased reverse bias shows a larger EE bandwidth, resulting from the smaller junction capacitance due to the extended intrinsic region. The other influence factor is the

device length. Intuitively speaking, a longer device will suffer a larger RF loss, which agrees well with the results in Fig. S9C. Noteworthy that different optical structure does not significantly affect the RF loss if their electrode lengths are at the same level, verified by the almost coincident curves between a 244 μm device ($N_p = 40, N_r = 10$) and another 249 μm device ($N_p = 20, N_r = 20$). For most of the devices whose lengths are under 300 μm , the EE bandwidth is well beyond 110 GHz, while in device D whose length is 489 μm , the RF transmission shows drastic attenuation, indicating that the microwave loss becomes the bottleneck that limits the EO bandwidth.

As for the EO bandwidth measurement, the S-parameter is tested by a VNA and its plugged-in light-wave component analyzer (LCA), with the setup illustrated in Fig. S10. The RF probing signal generated by the VNA is firstly mixed with the DC bias voltage through Bias-Tee and then loaded onto the modulator. The modulated signal is amplified by an erbium-doped fiber amplifier (EDFA) and injected into the LCA. LCA converts the optical signal into the electrical signal and then sends it back to the VNA for S-parameter analysis. Through the mentioned procedure, we obtain the EO bandwidth.

For the optimized device design, the 3 dB EO bandwidth of the fabricated modulator is around 110 GHz in Fig. 3D. In this measurement, the bias voltage of the PN junction was set to be 3.5 V, which is actually the same as that we perform large-signal measurements (e.g. eye diagram, etc.) for the consistency of the characterizations. Actually, the operation bandwidth can be even larger if we apply a different test condition. For example, by slightly increasing the reverse bias voltage to 4 V, the 3 dB bandwidth could further exceed 110 GHz as shown in Fig. S11, owing to

the decreasing of junction capacity at higher bias voltage.

9 The measurement of eye diagram and BER

The setup of eye diagram measurement is illustrated in Fig. S12. The optical signal in C-band is generated by the tunable laser. After passing a polarization controller to minimize the coupling loss, the optical signal is sent into the silicon modulator by a grating coupler. Here, the bit pattern generator (BPG) and the 2:1 multiplexer (MUX) are adopted to generate high-speed differential on-off keying (OOK) signal. Then the signal is amplified and combined with bias voltage together to load onto the modulator through RF probe. After the conversion from electrical signal to optical signal by the modulator, the modulated optical signal is amplified by the EDFA and recorded by using the digital serial analyzer (DSA, sampling oscilloscope) to demonstrate the eye diagram.

Thanks to the record-high EO bandwidth and wide optical operation window, the modulator is capable of supporting high-speed signal transmission at different wavelengths without applying any digital signal processing (DSP). The DSP-free eye diagrams of 100 Gbps OOK signal in a spectral window of 8 nm are shown in Fig. S13. The measured extinction ratio (ER) at 1548 nm, 1550 nm, 1552 nm, 1554 nm and 1556 nm is 3.03 dB, 3.15 dB, 3.13 dB, 3.14 dB, and 3.15 dB, respectively. It can be seen that the modulator possesses excellent and consistent performance across the operational passband of 8 nm, which is suitable for DWDM-based and microcomb-driven optical communication applications.

Specifically, the attributes including the OMA, SNR, ER from 100 Gbps DSP-free OOK eye

diagrams at different wavelengths are summarized in Table S1 that corresponds to the results in Fig. 4G, showing the designed device has good wavelength consistency across the tests.

In terms of the power consumption per bit, the PN junction model shows that the capacitance of the PN junctions is 1.5 pF/cm under the bias voltage of 3.5 V. Combining with the device length of 124 μm and the differential signals V_{pp} of 5 V, the power consumption per bit of the modulator for OOK ($E_b = 2 \times 1/4CV_{pp}^2$) operation is calculated to be 232.5 fJ/bit⁶⁴. Although the relatively large and differential drive voltage increases the power consumption, the ultra-compact footprint of the modulator makes the energy consumption per bit fairly acceptable.

Meanwhile, the fabricated modulator shows good temperature stability. Because the operation window is wide enough, we in principle do not need a TEC thermal control to maintain the temperature to a dedicated value as micro-ring modulators. We have tested the fabricated modulator at different operating temperatures from eye diagram measurement for 100 Gbps OOK signal (Fig. S14). The modulator's performance is highly uniform, the ERs are measured to be quite stable at 20 °C, 30 °C, and 40 °C which are 3.22 dB, 3.27 dB, and 3.11 dB, respectively.

Furthermore, we present the DSP-free eye diagrams of PAM-4 signal under a group of data rates of 52 Gbps, 106 Gbps, and 116 Gbps, respectively, as shown in Fig. S15, proving the proposed modulator is capable of high-order modulation. The measurement of the PAM-4 transmission of 116 Gbps (Fig. S15C) shows the quality of the eye diagram degrades to some extent, suggesting that a higher driving voltage and better linearity at high frequency are required for further increasing of the baud rate. We believe that the performance of high-order modulation can be optimized

by promoting the modulation depth, for instance, appropriately increasing the device length or slow light effect to increase the modulation efficiency.

As for the BER measurement, the same configuration but an arbitrary wave generator (AWG) for programmable signal generation is adopted for data transmitting, shown in Fig. S16. At the receiving end, a real-time oscilloscope is employed to capture the waveform, and the BER is off-line calculated based on FFE (Feed-Forward Equalization) and MLSE (Maximum-Likelihood Sequence Estimation).

In order to demonstrate the data transmission more quantitatively, the measured BER curves versus received optical power are illustrated in Fig. S17, under a variety of data rates at 70 Gbps, 84 Gbps, 98 Gbps and 112 Gbps. Specially, the scenarios of FFE and MLSE are shown and the adoption of MLSE will improve the BERs to a certain extent. Here, the BER at different wavelengths and different rates are summarized in Table S2 that corresponds to the results of Fig. 4H. The BERs can drop well below the hard-decision forward-error-coding (HD-FEC) threshold under 98 Gbps, and below the soft-decision forward-error-coding (SD-FEC) threshold when up to 112 Gbps.

Table S1: | **The measured indicators including OMA, SNR, ER of 100 Gbps**

Wavelength	OMA (mW)	SNR	ER
1548	2.50	3.72	3.03
1550	2.56	3.24	3.15
1552	2.59	3.43	3.13
1554	2.61	3.58	3.14
1556	2.51	3.40	3.15

Table S2: | **The measured BER at different wavelengths and rates**

Wavelength	70 Gbps	84 Gbps	98 Gbps	112 Gbps
1548	8.99×10^{-4}	1.11×10^{-2}	4.07×10^{-2}	8.47×10^{-2}
1550	1.86×10^{-5}	9.85×10^{-5}	2.17×10^{-3}	1.64×10^{-2}
1552	1.88×10^{-4}	1.87×10^{-3}	1.20×10^{-2}	3.22×10^{-2}
1554	3.27×10^{-4}	7.79×10^{-4}	8.45×10^{-3}	2.99×10^{-2}
1556	8.19×10^{-4}	1.58×10^{-3}	7.14×10^{-3}	4.34×10^{-2}

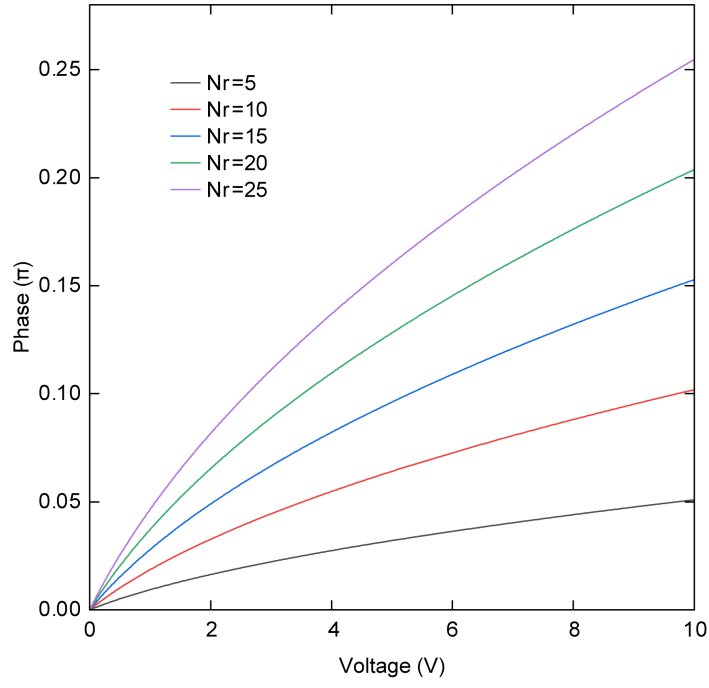


Fig. S1: | **The nonlinear dependence of modulation phase upon voltage.** Here we set $N_p = 20$ and apply a set of N_r from 5 to 25 for calculation, including $N_r = 10$ which is the optimal design that we test in the experiment. All the phase-vs-voltage curves show square-root dependency.

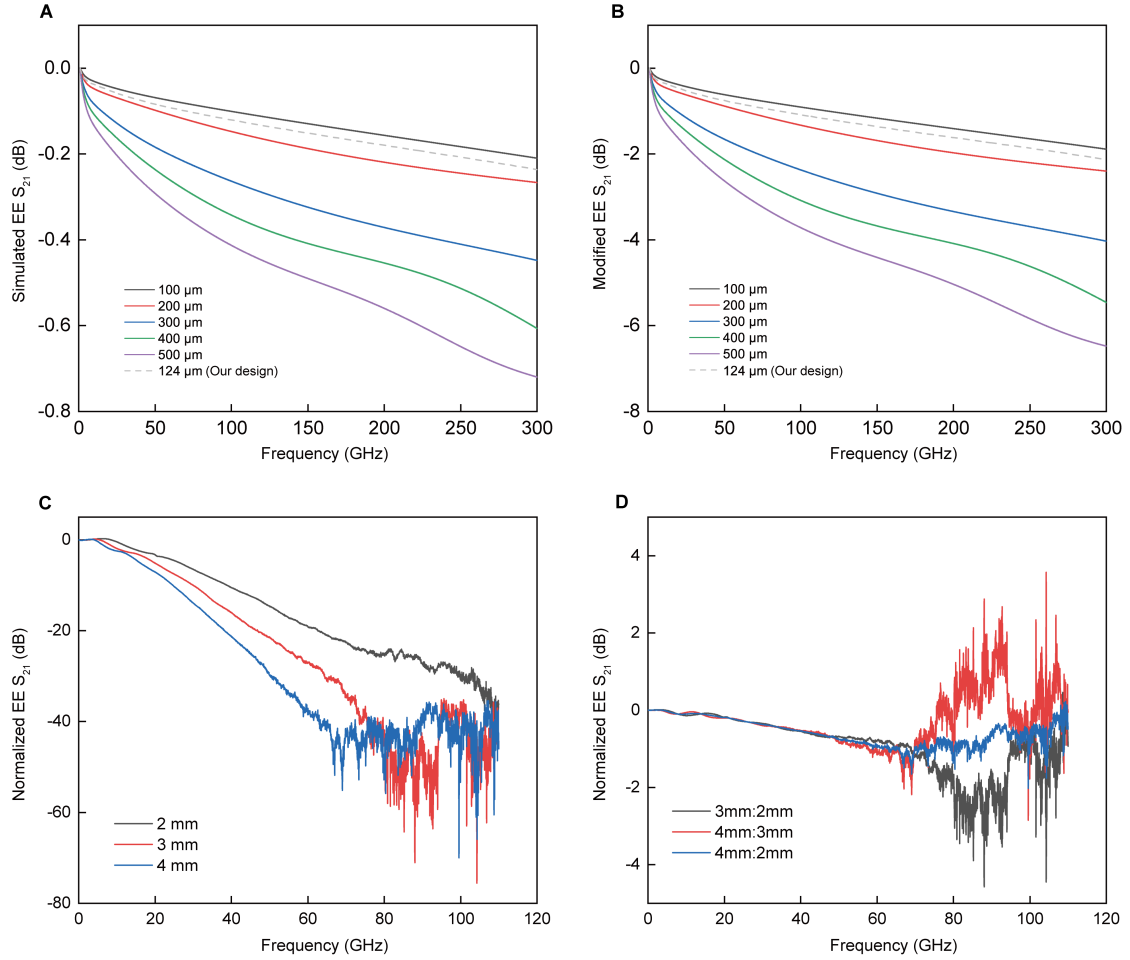


Fig. S2: | **The microwave electrodes loss with different length.** (A),(B) The electrical-to-electrical (EE) S_{21} responses of electrodes by using (A) the perfect metal model and (B) a model with experimentally measured loss taken into account. (C) Measured EE S_{21} responses of the silicon modulators with the designed electrode of 2 mm, 3 mm, and 4 mm. (D) The differential electrode losses are derived from (C) and normalized to a length of 100 μm .

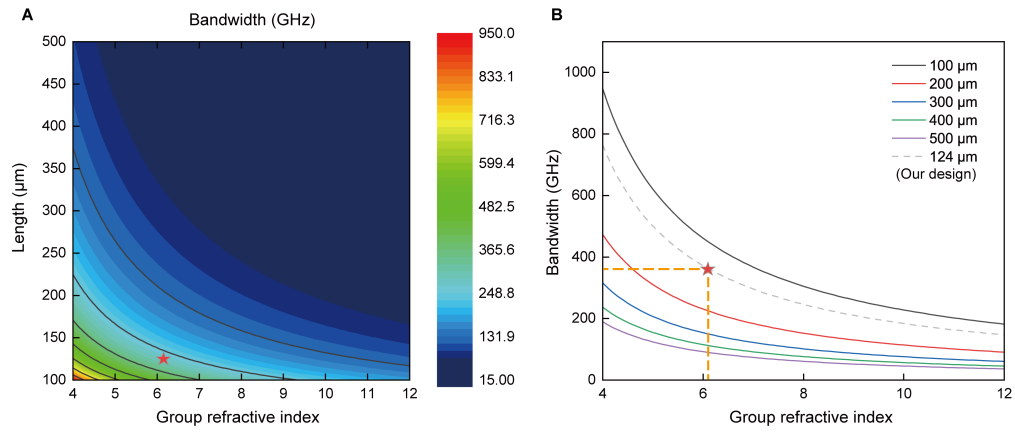


Fig. S3: | **The microwave-light-wave mismatching limited bandwidth.** (A) The contour and map of the calculated bandwidth on the electrode length and the group index of the lightwave. (B) The bandwidth dependency on the group index of the lightwave under different electrode lengths. The star mark shows our optimal design.

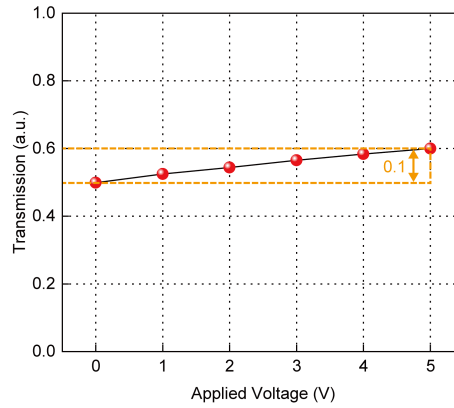


Fig. S4: | **The measured optical transmission extinction with the applied voltages.**

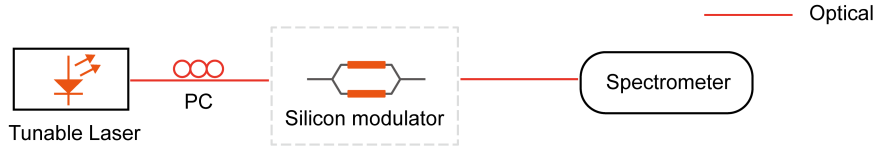


Fig. S5: | The experimental setup of spectra measurement.

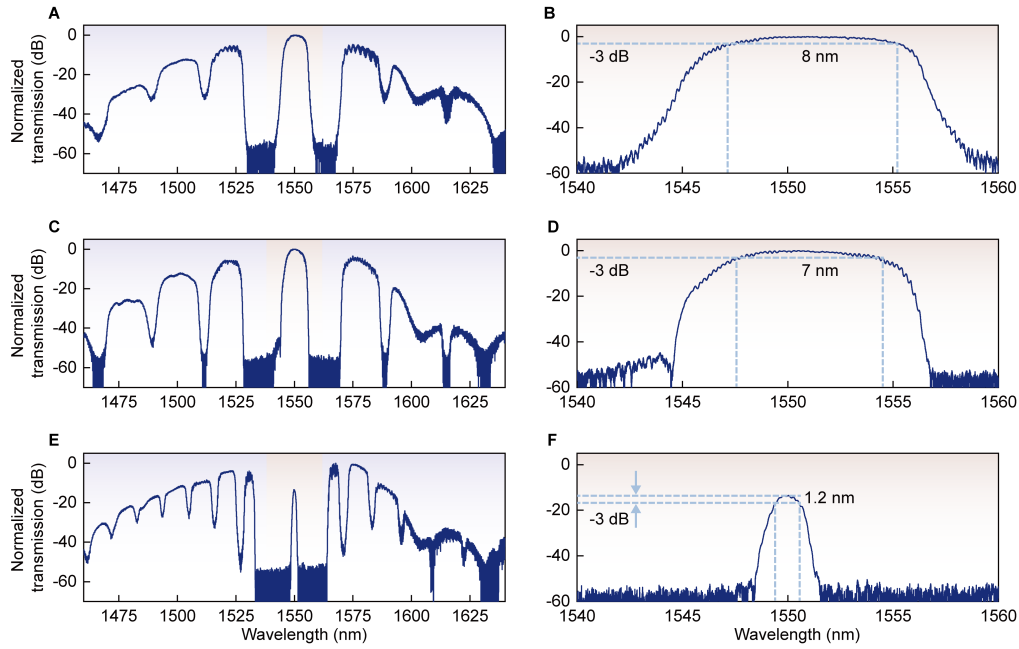


Fig. S6: | The measured transmission spectra of the modulators with different N_p and N_r .

(A) Device A ($N_p = 20, N_r = 10$) from 1460 nm to 1640 nm. (B) The detailed spectrum of Device A from 1540 nm to 1560 nm. (C) Device B ($N_p = 20, N_r = 20$) from 1460 nm to 1640 nm. (D) The detailed spectrum of Device B from 1540 nm to 1560 nm. (E) Device C ($N_p = 40, N_r = 10$) from 1460 nm to 1640 nm. (F) The detailed spectrum of Device C from 1540 nm to 1560 nm.

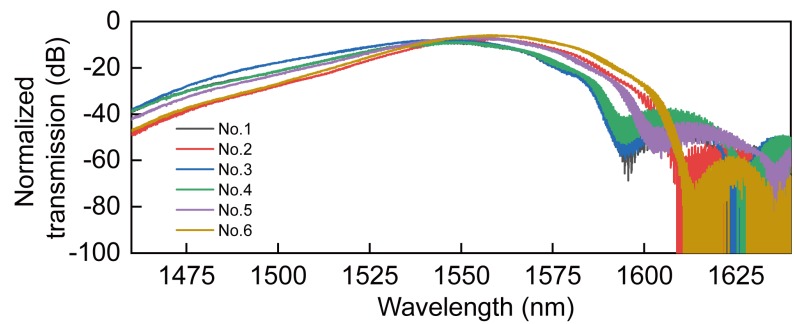


Fig. S7: | **The measured transmission spectra of grating couplers.**

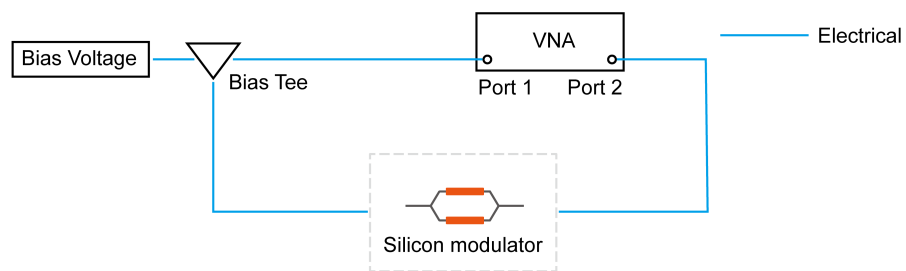


Fig. S8: | **The experimental setup of EE bandwidth measurement.**

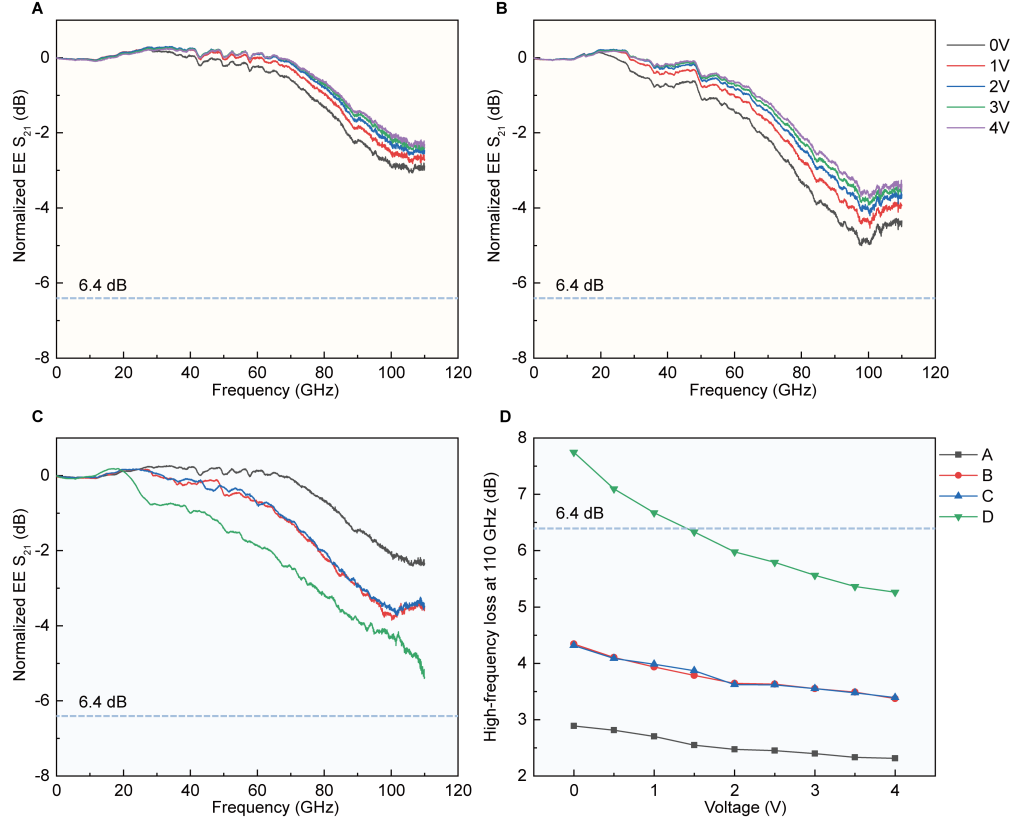


Fig. S9: | **The measured EE S_{21} responses for different modulator of Device A to D.** (A),(B).

The EE S_{21} responses under different bias voltages for (A) Device A, and (B) Device B. We found sufficiently short modulation arm is necessary in ensuring ultra-high bandwidth. (C) The EE S_{21} responses under 3 V bias voltage. (D) The measured microwave loss under different bias voltage for the Devices from A to D. Device A: ($N_p = 20$, $N_r = 10$, $127 \mu\text{m}$), Device B: ($N_p = 20$, $N_r = 120$, $249 \mu\text{m}$), Device C: ($N_p = 40$, $N_r = 10$, $249 \mu\text{m}$), Device D: ($N_p = 40$, $N_r = 20$, $489 \mu\text{m}$).

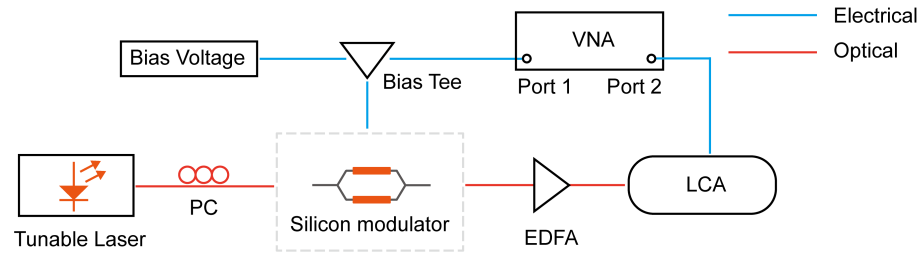


Fig. S10: | **The experimental setup of EO bandwidth measurement.**

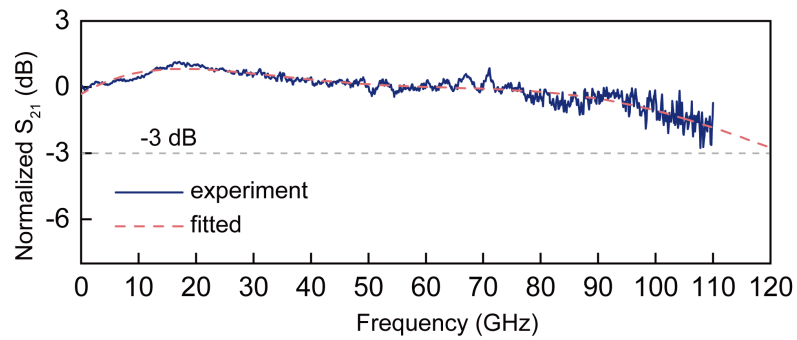


Fig. S11: | **The EO response under the bias voltage of 4 V.**

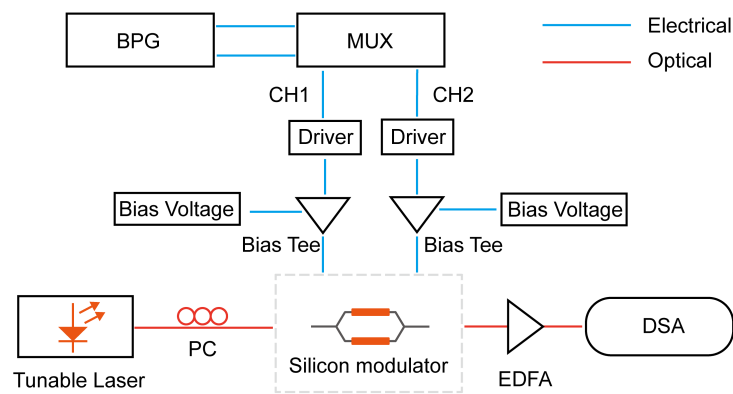


Fig. S12: | **The experimental setup of eye diagram measurement.**

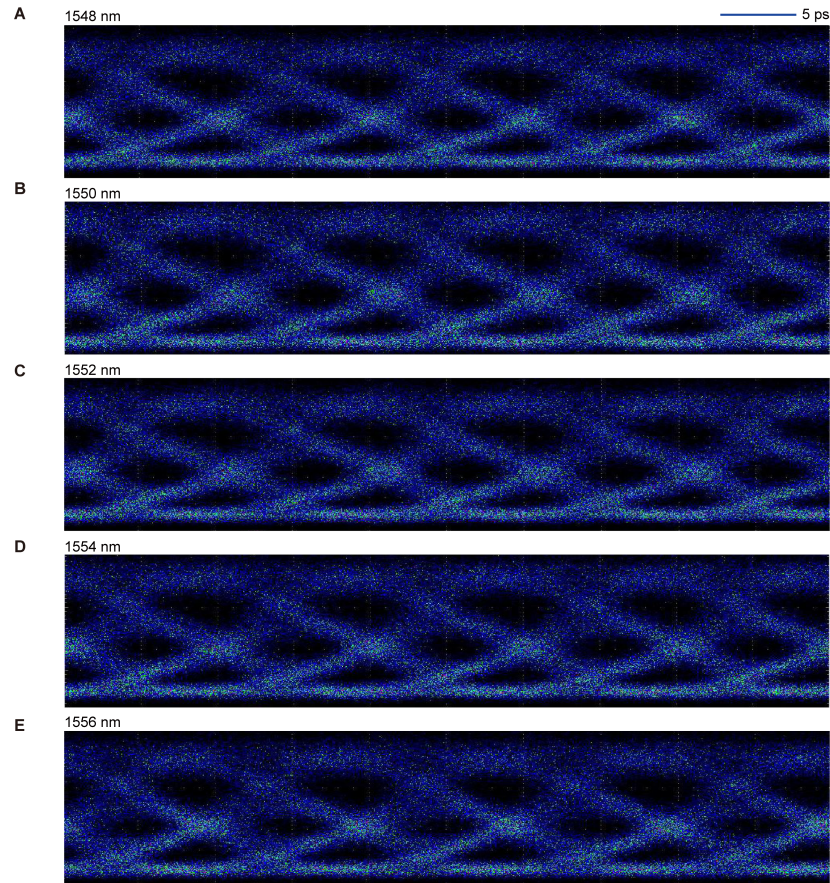


Fig. S13: | **The DSP-free eye diagrams of 100 Gbps OOK signal at different wavelength.** The measured extinction ratio (ER) at (A) 1548 nm, (B) 1550 nm, (C) 1552 nm, (D) 1554 nm and (E) 1556 nm are 3.03 dB, 3.15 dB, 3.13 dB, 3.14 dB and 3.15 dB, respectively.

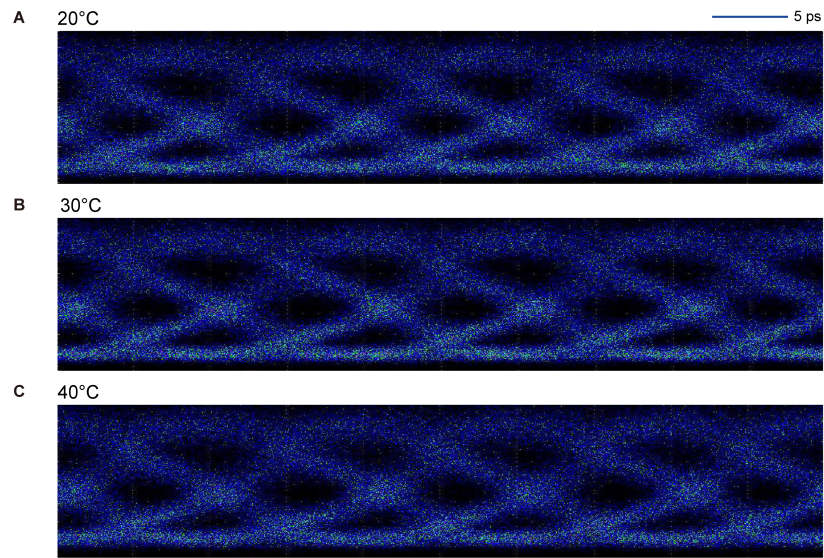


Fig. S14: | **The DSP-free eye diagrams of 100 Gbps OOK signal at different temperature.** The measured extinction ratio (ER) at (A) 20 °C, (B) 30 °C, (C) 40 °C are 3.22 dB, 3.27 dB and 3.11 dB, respectively.

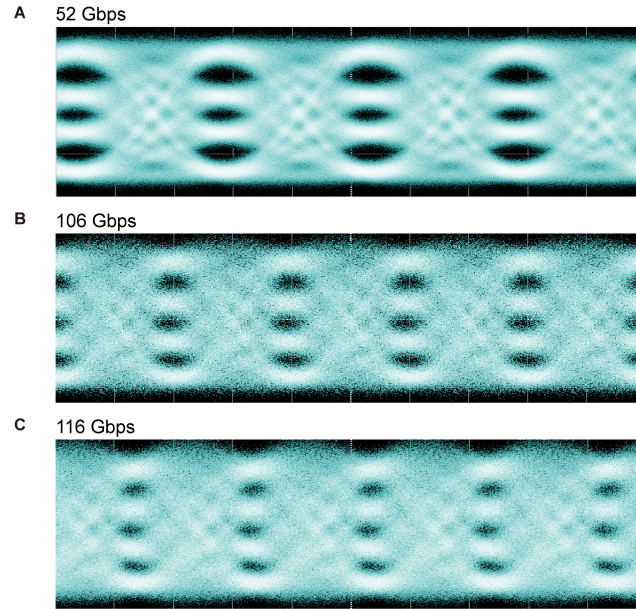


Fig. S15: | **The DSP-free eye diagrams of PAM-4 signal at different rates.** (A) 52 Gbps (B) 106 Gbps (C) 116 Gbps

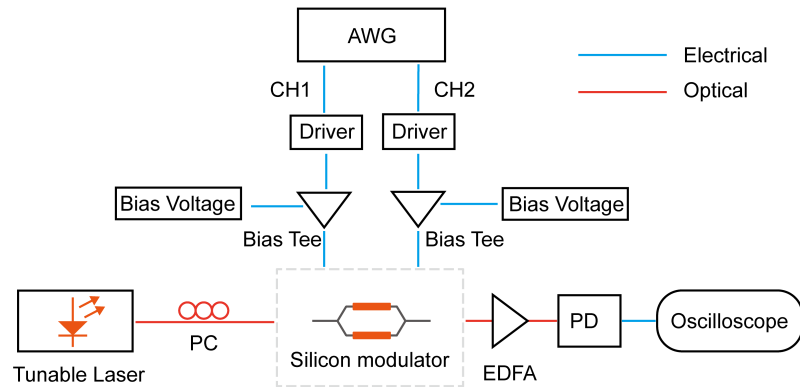


Fig. S16: | **The experimental setup of BER measurement.**

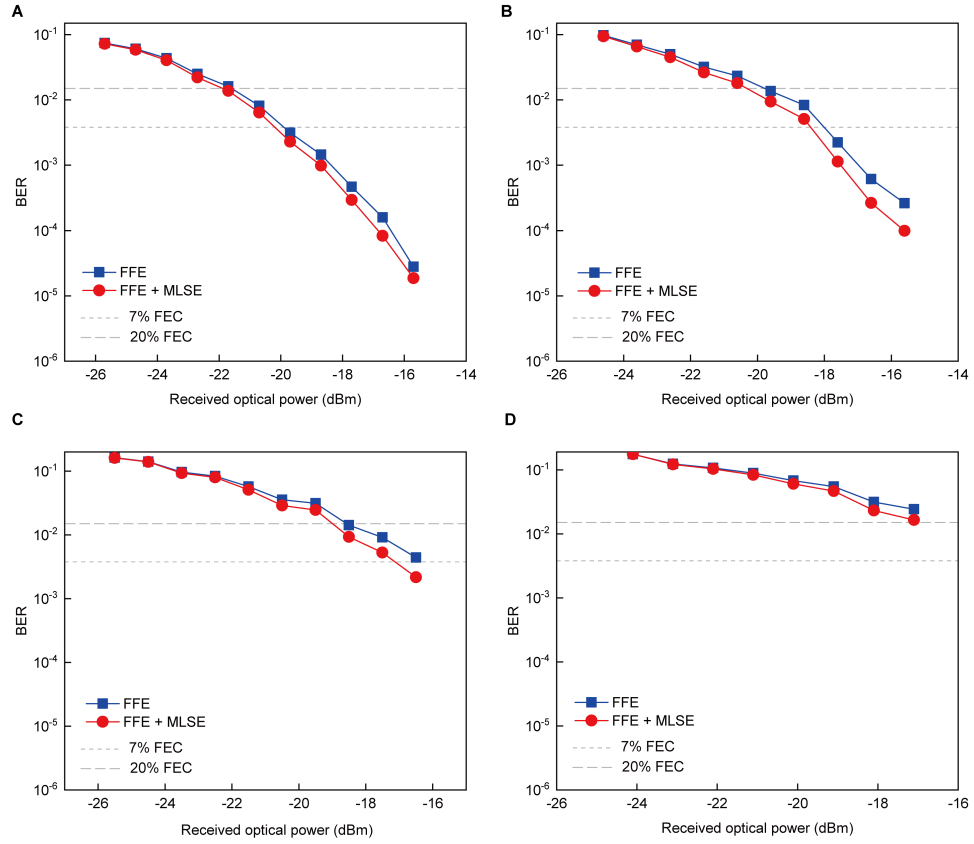


Fig. S17: | **The measured BER curves versus received optical power.** The BERs under a variety of data rates at (A) 70 Gbps, (B) 84 Gbps, (C) 98 Gbps and (D) 112 Gbps. The scenarios of FFE and MLSE are shown.